

Introduction

The success of your design solution relies heavily on your understanding of the problem you are addressing. Typically, you must satisfy widely varying demands and produce results that benefit numerous stakeholders (individuals, groups, and society as a whole). This exercise enables your team to demonstrate your collective understanding of the needs your solution addresses and the benefits it should deliver. You should gain some insight into the constraints on your problem and its solution.

Assignment

As a team, complete the assignment outlined on the following pages. Use the worksheets provided and prepare one response for your team. In your response, provide detail that demonstrates your understanding of *Solution Requirements and System Specification* for your project.

Grading

Your response will be graded on your ability to identify and explain stakeholder needs and corresponding expectations (or targeted requirements) for your design solution and on the quality of your writing. As appropriate, address requirements that relate to customers (or end users of the solution), business-related needs, limitations and potential of the technology, and societal/professional expectations.

Terminology

The term *specification* does not have a universally accepted meaning. Sometimes the plural term ***specifications*** refers to the collection of characteristics of an already designed product. In this case, each item in a list of specifications is a characteristic that an end user can expect from the product. It is common for a consumer product to have such a list of *specifications* for the product. For example, see the listed “specifications” for the Sennheiser SpeechLine DW (Digital Wireless) product beginning on page 3 of 7 at [SP_1027_v1_0_SL_Digital_Wireless_System_Specification_EN.pdf \(sennheiser.com\)](#)

Alternatively, the term ***specification*** sometimes refers to a collection of *requirements*, often stated within a document referred to as the specification for an item or system. This specification document is typically written prior to an item or system being developed, to state the expectations and define the performance requirements. Sometimes such a specification is referred to as a *requirements specification*. For example, see the “Architect’s Specification” for the Sennheiser SpeechLine DW (Digital Wireless) product beginning on page 1 of 7 at [SP_1027_v1_0_SL_Digital_Wireless_System_Specification_EN.pdf \(sennheiser.com\)](#)

In EE Senior Design, project teams develop specifications for their overall system and develop specifications for each of their subsystems. None of these items currently exist so you are setting expectations for their performance. Therefore, the term *specification* will most often be used to refer to a collection of requirements as described in the previous paragraph. However, with concurrence from a team’s advisor, each team can use the term in whatever manner best suits the team.

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Team: _____ Date: _____

Part A: Problem Statement

The problem statement is one sentence (ideally) that clearly and concisely defines the problem to be solved. It is stated in the language of the customer. Needs can be described qualitatively. The problem statement should be consistent with the requirements of your customer. The solution envisioned and the benefits offered by your intended solution were described in your earlier Problem Statement submittal. Include any updates, additional refinements and details in the remainder of this submission.

Part B: Stakeholder Needs

Identify the most critical needs to be satisfied by your project solution. In the table below:

- (a) Identify your most important customers, users, or other stakeholders (or groups), and
- (b) List the needs they see as most important to the success of your project.

A **stakeholder** is any person or group that has an interest, investment, or share in the project. Examples include investors, designers, management, end users, manufacturers, technical specialists, sales and marketing people, suppliers, government, employees, interest groups, or society in general.

The best design solutions satisfy needs of all key stakeholders. The purpose of this exercise is to clarify your understanding of the most important stakeholders and their needs regarding your design solution.

Example Stakeholder: User of the new product

The solution must:

- Perform better than existing products of similar function
- Operate remotely in an un-tethered manner
- Weigh less than 5 lbs

Stakeholder #1: Homeowners / Smart-Home Users

The solution must:

- Be significantly more affordable than comparable commercial smart-home sensors.
- Reliably detect the intended condition, such as motion, occupancy, ambient light level, or a door opening.
- Integrate easily with the user's existing smart-home hub and lighting system.
- Require minimal technical knowledge to install, configure, and use.
- Respond quickly enough that automatic lighting feels natural to the user.
- Operate reliably without frequent resets, communication failures, or false detections.
- Require little maintenance and, if battery powered, provide reasonable battery life.

Stakeholder #2: Product Developer / Manufacturer

The solution must:

- Use low-cost and readily available electronic components.
- Be inexpensive enough to manufacture that the final sensor can compete with existing smart-home products.
- Use a common hardware/software platform when possible so multiple sensor types can be developed without redesigning the entire system.
- Be straightforward to manufacture, assemble, program, and test.

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- Be modular enough that additional sensor types could be developed in the future.
Stakeholder #3:
The solution must:

Part C: Existing Solutions

Most designs are variations or derivatives from previous designs. Provided intellectual property laws and ethical standards are followed, analyzing the requirements satisfied by a similar product or system is not only appropriate, but may be necessary for success. Identify solutions (i.e., products, systems, procedures, etc.) that already exist, that solve the problem statement, and that meet expectations or requirements.

Indicate the extent to which the existing solutions solve the problem. Compare published specifications and expected performance of the existing solutions with the requirements listed in your system specification. Provide cost data where available. For the existing solutions you identify, when available, include published specifications (limit to a few pages) in your submittal or provide weblinks to those specifications. Identify any engineering standards that the existing solutions comply with.

Part D: Solution Requirements – System Specification

Define the most important technical, business, regulatory (e.g. engineering standards, protocols), ethical, social, environmental, health and safety or other requirements for your project solution. Be as specific as you can. These requirements can be functional and technical. These requirements will be important later because the System Verification (at the end of second term) will check to see if your design meets the requirements stated in your system specification.

For each:

- (a) Give a brief definition or description of the requirement.
- (b) Specify an expected value (or range of values) or a desired condition of operation for satisfying the requirement.
- (c) State the importance of satisfying this requirement for you to achieve a successful solution.
- (d) Note whether you intend to test or verify in some other manner compliance with the requirement and/or if you intend to certify compliance some other way, e.g., manufacturer datasheets/third-party testing.

<i>Description of Requirement</i>	<i>Expected Value</i>	<i>Importance*</i>	<i>Compliance Verification Method</i>
<i>Example 1: Positioning accuracy for the reader head</i>	<i>+/- 0.02 mm</i>	<i>H</i>	<i>Datasheet</i>
<i>Example 2: Deformation (dent size) on enclosure after a 3-ft drop</i>	<i><= 10 sq. in.</i>	<i>M</i>	<i>Measure</i>
<i>Example 3: Review by Acme Safety Inspection Laboratories</i>	<i>Approval level 1</i>	<i>E</i>	<i>External</i>
<i>Example 4: Allows any user in the network to initiate a 3-way call</i>	<i>Functions as stated</i>	<i>H</i>	<i>Demonstration</i>
<i>Example 5: Power consumption from power source</i>	<i><= 12 Watts</i>	<i>M</i>	<i>Test/Measure</i>
<i>Example 6: Power-up time from power-on to fully operational state</i>	<i><= 40 seconds</i>	<i>L</i>	<i>Measure</i>

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<i>Example 7: Depth, width and height</i>	<i><= (9",14", 4")</i>	<i>M</i>	<i>Measure</i>
<i>Example 8: Input voltage to power the system</i>	<i>12VDC +/- 10%</i>	<i>H</i>	<i>Test/Measure</i>
<i>Example 9: System notifies administrator of system fault via automated text message</i>	<i>Functions as stated</i>	<i>E</i>	<i>Demonstration</i>

* Importance of the requirement: E = essential, H = high, M = moderate, L = low

In some cases, as in Examples 4 and 9 above, certain requirements are better described qualitatively in the form of a functional description of the solution. As necessary, include functional requirements describing important attributes and characteristics of the solution such as the system behavior, how a user will interact with the system, and scalability.

Grading rubrics

	Low	Medium	High
Requirements and existing solutions	Some marginally relevant existing solutions identified	Relevant existing solutions identified and analyzed	Appropriate new technology identified
	Some customer needs are converted to requirements	Most customer needs are converted to requirements	All customer needs are converted to requirements
	Inappropriate or no physical properties and/or operational conditions specified.	Most appropriate physical properties and/or operating conditions specified	All appropriate physical properties and/or operating conditions specified
	Power requirements not specified	Power requirements mostly specified	Power requirements fully specified
	Requirements difficult or impossible to verify	Most requirements may be verified; some have uncertain verification	All requirements may be verified
	Current standards are not fully considered	Many current and relevant standards are incorporated in the project expectations	An impressive set of current and relevant standards are incorporated in the project expectations
	Requirements/constraints that would be considered in industry are not examined	Many requirements/constraints that would be considered in industry are included	Fully understands all vital customer needs including requirements/constraints that would be considered in industry

	Low	Medium	High
Customer	Does not identify principal customers	Identifies principal customers	Identifies primary and secondary customers
	Identifies only most obvious customer needs	Identifies most important customer needs	Identifies all vital needs for competitive advantage
	Does not show understanding of customer needs	Shows moderate understanding of customer needs	Fully understands all vital customer needs

	Low	Medium	High
Business	Does not identify financial stakeholders or their interests	Identifies principal financial stakeholders and some interests	Knows financial stakeholders, their desires and concerns
	Unaware of business-technical interactions	Aware that technical and financial issues interact	Understands business-technical trade-offs
	Has no sense of time constraints for success	Sees costs associated with time	Understands business plan and market strategies for product success

	Low	Medium	High
Technology	Not aware of technology-driven limitations or barriers	Aware of obvious limitations and barriers for the technology	Understands relevant technological limitations and barriers
	Sees little opportunity for the technology	Sees some opportunity to advance the technology	Sees opportunities for innovation and advancement

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	Unaware of potential for obtaining valuable intellectual property	Recognizes some potential for intellectual property	Understands potential value and issues surrounding intellectual property
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	Low	Medium	High
Society	Not aware of relevant standards, codes, regulations	Aware of most important standards, codes, regulations	Clearly understands relevant standards, codes, regulations
	Only cursory attention to environmental or societal impacts	Shows concern for environmental and societal impacts	Advocates for people, environment, and society in general
	Appears irresponsible	Appears responsible	Highly responsible

Category	Below	Adequate	Above	Comments
Problem Statement appropriately updated	☐	☐	☐	
Stakeholders: Sufficiently complete set of them	☐	☐	☐	
Stakeholders: Include appropriate principal end-user needs	☐	☐	☐	
Stakeholders: Include business, societal, other stakeholders, and appropriate needs	☐	☐	☐	
Existing solutions: Sufficient number found	☐	☐	☐	
Existing solutions: Including shortcomings	☐	☐	☐	
Existing solutions: Inclusion of their Requirements/Specs/Standards and Cost (if available)	☐	☐	☐	
All customer needs converted to requirements	☐		☐	
Physical requirements fully specified (i.e. size, weight, etc)	☐	☐	☐	
Power requirements fully specified	☐	☐	☐	
Functional operational description	☐	☐	☐	
Safety requirements fully specified	☐	☐	☐	
Relevant standards addressed	☐	☐	☐	
Writing quality	☐	☐	☐	

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Solution Requirements and Specifications

Smart Home Manager

09/20/26

Miles Bailey-Waltz

Jackson Duty

Megan Ruffing

Lily Schellenbach

Atifa Tariq

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Part A: Problem Statement

Homeowners need a reliable and easy way to automatically control their lighting based on room conditions and activity without requiring complicated installation, or major changes to their existing smart-home systems.

Part B: Stakeholder Needs

Stakeholder #1: Homeowners / Smart-Home Users
The solution must: • Be significantly more affordable than comparable commercial smart-home sensors. • Reliably detect the intended condition, such as motion, occupancy, ambient light level, or a door opening. • Integrate easily with the user's existing smart-home hub and lighting system. • Require minimal technical knowledge to install, configure, and use. • Respond quickly enough that automatic lighting feels natural to the user. • Operate reliably without frequent resets, communication failures, or false detections. • Require little maintenance and, if battery powered, provide reasonable battery life.
Stakeholder #2: Product Developer / Manufacturer
The solution must: • Use low-cost and readily available electronic components. • Be inexpensive enough to manufacture that the final sensor can compete with existing smart-home products. • Use a common hardware/software platform when possible so multiple sensor types can be developed without redesigning the entire system. • Be straightforward to manufacture, assemble, program, and test. • Be modular enough that additional sensor types could be developed in the future.
Stakeholder #3: Building Developers/ Home Contractors
The solution must: • Be easy to install in both new construction and existing homes. • Use standard wiring, mounting methods, and commonly available components. • Require minimal additional labor, specialized tools, or technical training. • Meet applicable residential electrical, safety, and building-code requirements. • Be flexible enough to work in rooms with different sizes, layouts, and lighting configurations. • Include clear installation, setup, and troubleshooting instructions. • Be affordable enough to install throughout a home without significantly increasing construction costs. • Provide measurable energy savings that building developers and contractors can promote to customers as a way to reduce long-term electricity costs.

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Part C: Existing Solutions

Matter has existing sensors, but because it was designed to integrate existing smart home applications, most of its existing sensors are expensive for their application because they are designed to be stand alone systems that integrate into Matter. So we will be using the Matter protocol that already exists that integrates these devices together.

If a person who already has a thermostat wants to measure the temperature downstairs using Matter, the cheapest standalone temperature sensor is the Switchbot Room Thermometer, which retails on amazon for \$13.99

(https://www.amazon.com/dp/B07L4QNZVF?ref=emc_p_m_5_i_atc&th=1). This is a combination temperature and humidity sensor, and while it is inexpensive, it uses an LCD screen. This means that there is room for the device to be smaller and more inexpensive than it currently is. Switchbot is the least expensive standalone temperature sensor by a wide margin. The switch bot complies with a class B digital device pursuant to part 15 of FCC Rules (<https://m.media-amazon.com/images/I/A10TJH7ts3L.pdf>) When we build something along these lines we will also have to comply to the Class B since that is the standards for RF with residential. Here is a link to what those specifications are for Class B (<https://compliancetesting.com/fcc-part-15-class-a-class-b-limits/>). There are other solutions as well but the solution we want to create is mostly working to make these sensors more affordable so I analyzed the cheapest one and we want to be able to perform the same as this current one.

Part D: Solution Requirements – System Specification

Requirements	Description	Expected value/ Desired Condition	Level of importance	Testing
Automated lighting control	Be able to control the lighting in a room based on the light level	Dim or brighten lights when ambient light exceeds or falls below a certain threshold.	Very Important – This is an easily demonstratable milestone for automation	Sensor will be tested in an environment with a smart hub and a controlled light to see if the light dims when covered.
Matter Hub Compatible	Compatible with existing smart hub protocol.	Sensors must comply with the matter 1.6 core specification document.	Very Important – The main desired outcome of this project is to make more affordable sensors that use the matter protocol.	Must comply with this Specification: https://csa-iot.org/wp-content/uploads/2026/06/23-27349-011_Matter-1.6-Core-Specification.pdf
Light Energy Usage	The main goal of this project is to create a smart lightning system that will reduce unnecessary power usage, meaning when there	The lights would remain off for as long as there is enough lightning in the room or it is not the appropriate time of day.	Important – One of the many reason, users want smart devices is so they no longer must monitor their power usage.	Testing will be done by gathering energy data from this device and comparing it to energy spent from manual lights in a day.

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	is no need for lights, the lights will remain off.			
Sensor Energy Usage	The sensor should be able to remain operational with minimal resident maintenance on 2 AA batteries.	The sensor should be designed to be powered off of 2 AA batteries, which have 6-8 watt hours of energy total. If the need to change the batteries is too frequent, it will not be useful in residential applications. Ideally, the sensor should use 35 mW/h so that it can operate for a full week without needing the batteries to be replaced, although ideally it would be even longer.	Very important – End users will not want a product that has a costly upkeep.	Altium provides utilities for power consumption, which will be used during the board prototyping phase. After a design has been finalized, the power consumption will be measured using two standard AA batteries. Tests should also be conducted using rechargeable batteries, as users may end up using them.
Board Weathering	The end user may want to place their board outside in the elements, and while the boards are designed to be affordable, we do not want them to be disposable.	Ideally, the board would have a case that complies with IP32(protection from objects >2.5mm and water from being tilted up to 15 degrees) at a minimum, but ideally IP54(resistant to incident dust and splashing water from any direction).	Somewhat important – users may want to use them outside, but a functional design should be prioritized.	The board can be tested by being sent off for testing, but it's not a priority that it be done. Functional design comes first.

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Unit Cost	In order to be effective, the product must cost less than the existing solutions	This is typically generous, but for each we should aim to stay underneath the next most expensive product made for Matter.	Very Important – Users will only consider the product over other alternatives if it is cheaper, and forgive some less than desirable behavior if it does.	Keep a bill of materials for each sensor and check against the ever updating database of matter devices here: https://www.matteralpha.com/categories?page=1
Size	The board has to have a minimum size that holds two AA batteries.	Minimum of 28 mm x 50 mm	Not very important – As long as it's larger than the batteries, it meets spec	Make sure the Altium designed board isn't smaller than 28x50(mm), and that there is a place for the batteries to go.

